

PROJECT	Credit Card Approval Prediction	TEAM	AI/ML Program
DOCUMENT	Coding & Solution	DATE	July 2026

Coding & Solution

Data Preprocessing (Summary)

- Dropped OCCUPATION_TYPE column (~30% missing values)
- Removed duplicate applicant records by ID
- Converted DAYS_BIRTH and DAYS_EMPLOYED to positive values
- Handled the 365243 placeholder value for unemployed/pensioner applicants
- Converted the multi-class credit STATUS column into a binary TARGET (0 = approved, 1 = rejected)
- Aggregated credit history per applicant and merged with applicant details on ID
- Engineered a FAMILY_SIZE feature from CNT_CHILDREN + CNT_FAM_MEMBERS
- Label-encoded all categorical columns

Model Training (Summary)

Four classification algorithms were implemented: Logistic Regression, Decision Tree (`d_tree()`), Random Forest (`random_forest()`), and XGBoost. Each model was trained on an 80/20 stratified train-test split and evaluated using a confusion matrix and classification report.

Key Code Snippet - Final Model Selection

```
rf_model = RandomForestClassifier(random_state=42, class_weight='balanced')  
  
rf_model.fit(X_train, y_train)  
  
pickle.dump(rf_model, open('model.pkl', 'wb'))
```

Application Integration

The Flask application (`app.py`) loads `model.pkl`, encodes incoming form submissions using the exact category mappings derived from training, converts user-friendly inputs (age, years employed) into the model's expected DAYS_BIRTH/DAYS_EMPLOYED format, and returns a rendered prediction result.